

Energy-stable WENO: what the three papers establish and what EXHALE uses

Abstract

The energy-stable WENO (ESWENO) schemes of Yamaleev & Carpenter (2009a,b) and their finite-domain closures (Fisher et al. , 2011) add to a conventional WENO flux a nonlinear artificial dissipation term built from the WENO weights themselves, and prove that the resulting finite-difference operator is L_2 -stable for linear hyperbolic equations with continuous or discontinuous data, with no total-variation assumption and no time-step restriction beyond the usual one. The same papers introduce new weight functions that hold the design order at smooth extrema with any number of vanishing derivatives, and the 2011 paper supplies summation-by-parts boundary closures so the estimate holds on a finite domain. EXHALE's WENO3 reconstruction already uses the 2009a weight functions and their $\epsilon = O(\Delta \xi^2)$ floor; it does not compute the dissipation term, so it inherits the accuracy result and not the energy estimate. Three properties of the code keep the estimate from carrying over as published: a finite-volume reconstruction of the primitive state on a stretched grid, a component-wise rather than characteristic reconstruction, and the nonlinear Euler system, for which the papers themselves give no proof. This memo records what each paper proves, at which equation, and what would have to change for the estimate to apply.

1. The question

The question asked on 2026-09-12 was what ESWENO gains over conventional WENO. The three papers are in ../references/ (the files Yamaleev_2009JCP_228_3025.pdf, Yamaleev_2009JCP_228_4248.pdf and Fisher_2011JCP_230_3727.pdf) and every statement below was read in those published versions; equation numbers are theirs. What the code does is read from src/modules/states/Reconstruction.f90, whose WENO3 branch carries its own account of the same papers.

2. The third-order scheme (Yamaleev & Carpenter , 2009a)

2.1 The construction

The scalar wave equation $u_t + f_x = 0$, $f = au$, $a \geq 0$, is discretized on a uniform periodic grid as (their Eq. 9)

$$\frac{\partial \mathbf{u}}{\partial t} + P^{-1} Q \mathbf{f} = -P^{-1} D_1^T S D_1 \mathbf{f}, \quad (1)$$

where $P^{-1} Q$ is the skew-symmetric part of the conventional third-order WENO derivative matrix, D_1 the two-point difference operator, and S a symmetric matrix that is a nonlinear function of the solution through the WENO weights. The left side is the WENO scheme; the right side is the addition. Multiplying by $\mathbf{u}^T P$ gives (their Eq. 11)

$$\frac{d}{dt} \|\mathbf{u}\|_P^2 + a \mathbf{u}^T (Q + Q^T) \mathbf{u} = -a (D_1 \mathbf{u})^T (S + S^T) D_1 \mathbf{u}, \quad (2)$$

so the energy cannot grow if $Q + Q^T$ vanishes in the periodic interior and S is positive semidefinite. Their S (Eqs. 35 to 37) is

$$S = S_1 + D_1^T S_2 D_1, \quad s_{jj}^1 = \mu_j - \frac{1}{8} (w_{j+1/2}^1 - w_{j-1/2}^1), \quad s_{jj}^2 = \frac{1}{4} w_{j-1/2}^1, \quad (3)$$

$$\mu_j = \frac{1}{8} \sqrt{(w_{j+1/2}^1 - w_{j-1/2}^1)^2 + \delta^2}, \quad (4)$$

with w^1 the weight of the upwind stencil and δ a small positive parameter that keeps μ_j a C^1 function of the solution. The proof of positive semidefiniteness (their Section 4.1) uses only two properties of the weights, their Eq. 23: $0 \leq w^r \leq 1$ and $w^0 + w^1 = 1$. It holds for any weight functions with those properties, continuous or discontinuous data alike (their Remark 6), and fails if a weight is negative (Remark 7). The stencil is the five points of conventional WENO3 (Remark 4).

The dissipation term is third order on smooth data, so the scheme keeps the design order there; it is what makes the operator's symmetric part negative semidefinite. The papers state as the distinctive feature that the symmetric part of the conventional WENO operator can have positive eigenvalues, so conventional WENO “may become locally unstable” (their Section 7), and that no energy estimate is available for it.

2.2 The new weight functions

Independently of the dissipation term, the paper replaces the Jiang & Shu (1996) weights $\alpha_r = d_r/(\epsilon + \beta_r)^2$ by (their Eqs. 18, 21, 22)

$$w^r = \frac{\alpha_r}{\alpha_0 + \alpha_1}, \quad \alpha_r = d_r \left(1 + \frac{\tau}{\epsilon + \beta_r} \right), \quad \tau = (u_{j+1} - 2u_j + u_{j-1})^2, \quad (5)$$

with $\beta_0 = (u_{j+1} - u_j)^2$, $\beta_1 = (u_j - u_{j-1})^2$, $d_0 = 2/3$, $d_1 = 1/3$. On smooth data $w^r = d_r + O(\Delta x^2)$ (their Eq. 24) with no assumption on the number of vanishing derivatives, provided $\epsilon \geq O(\Delta x^2)$ (Eq. 51); with $\epsilon = 0$ the ratio τ/β_r is $O(1)$ at a smooth extremum and both WENO and ESWENO drop to second order there (Eq. 57), which is what the Jiang–Shu weights do on practical grids (Eq. 58). Nullifying the stencil astride a discontinuity requires $\epsilon \leq O(\Delta x^2)$ (Eq. 61), so $\epsilon = O(\Delta x^2)$ (Eq. 62), made scale-invariant with the computational spacing $\Delta \xi = 1/J$ (Eq. 64) and given the solution scale by

$$\epsilon = \max(\|u_0^2\|, \|(u_0)_\xi^2\|) \Delta \xi^2, \quad \delta = \Delta \xi^2 \quad (6)$$

(Eqs. 65, 66; L_1 norms of the initial condition, evaluated once). The paper says explicitly that “these weights can also be used with the conventional third-order WENO scheme” (Section 3.1): the accuracy result and the energy estimate are separable, and only the latter needs the dissipation term.

2.3 Scope of the proof

The estimate is proven for the scalar equation and, in Section 5, for a system of constant-coefficient linear hyperbolic equations in *characteristic* form, “while it is not clear how to obtain a similar energy estimate when the system of equations is discretized in the component-wise fashion.” For the Euler equations the paper is explicit (Section 6.2): “No stability proof is currently available for the ESWENO scheme for the nonlinear Euler equations and the above characteristic decomposition which is done locally at each point.” The nonlinear tests (quasi-one-dimensional nozzle, subsonic and transonic; time-dependent problems with strong shocks and contacts) are reported as stable and essentially non-oscillatory, with a local Roe-averaged characteristic decomposition and global Lax–Friedrichs splitting, but as observations.

3. Arbitrary order (Yamaleev & Carpenter , 2009b)

The second paper turns the construction into a procedure for any order: decompose the symmetric part of the WENO operator into a sum $\sum_{i=0}^s [D_1^i] \Lambda_i [D_1^i]^T$ of nested two-point differences with diagonal Λ_i (their Lemma 1 and Appendix), replace each diagonal entry by its smoothly positive part $\frac{1}{2}(\sqrt{\lambda^2 + \delta_i^2} - \lambda)$ (Eq. 23), and add the resulting positive semidefinite operator (Eq. 24) as the dissipation term. Fifth-order upwind-biased schemes as a one-parameter family, a sixth-order central scheme, and schemes to eighth order are given, with weight functions of the form (5) and constraints on their parameters that keep design order with any number of vanishing derivatives (their Eq. 58 and the floor of Eq. 79 are the higher-order counterparts of the 2009a forms).

The measured cost (their Table 1, linear wave equation, 101 points, five periods): the third-, fifth- and sixth-order ESWENO schemes take 15 to 20 percent more CPU time than the WENO schemes they are built on, the fourth-order one somewhat more; no attempt at optimization was made. The stencil width is unchanged.

4. Boundary closures (Fisher et al. , 2011)

The 2009 estimates are for periodic domains. The 2011 paper builds closures for the fourth-order ESWENO scheme that satisfy the summation-by-parts convention, $D = P^{-1}[Q + R]$ with $Q + Q^T = \text{diag}[-1, 0, \dots, 0, 1]$ and R symmetric positive semidefinite (their Eqs. 4, 5), so that with a simultaneous-approximation-term boundary condition of strength $\sigma \leq -1/2$ the semi-discrete energy is bounded (Theorem 1). The finding that makes it possible is that the flux points near a boundary must be *nonuniform* and derived from the norm P (their Section 3.3), which is what lets accuracy, the SBP convention and the WENO stencil biasing hold at once. Second-order closures stable in a diagonal norm and third-order closures stable in a block norm give global third and fourth order. Test cases (their Section 7): the linear wave equation, steady quasi-one-dimensional nozzle flow with the nonlinear Euler equations, an unsteady two-dimensional Euler vortex, and a supersonic hydrogen–air mixing layer with the Navier–Stokes equations; an eigenvalue study shows the discrete operator stable for smooth and discontinuous data. The appendices give the near-wall smoothness indicators, a pseudo-code for the Euler equations, and the interpolation weights and dissipation coefficients of both closures.

5. What EXHALE uses today

The WENO3 branch of `Reconstruction.f90` builds its smoothness factors from Eq. (5): $\alpha_r = d_r(1 + \tau/(\epsilon + \beta_r))$ with τ the squared second difference, β_r the squared one-sided differences, $d_0 = 2/3$, $d_1 = 1/3$, and the floor $\epsilon = dr_j^2$, the local cell width squared, which carries the $O(\Delta x^2)$ scaling of Eqs. 62 to 64 with the local width standing in for the uniform $\Delta \xi$ and without the solution scale of Eq. 65. The code’s own note measures what that choice costs: nothing in a smooth region (the face error and the convergence rate on the production grid are flat for $\epsilon = (\kappa dr_j)^2$ over κ from 10^{-1} to 10^3 , $L_\infty 2.99 \times 10^{-8}$ at $N = 800$, rate 2.991, from the `weno3_reconstruction_order` test), and something in the stencil biasing, where on the WASP-121b reference output β_r/ϵ runs from 8.4×10^3 at the base to 2.2×10^{-6} at the top and a third of the cells sit below 10^{-2} , with no biasing left. So the code has the accuracy result of the 2009a weights (Eq. 24), the design order at smooth extrema included.

It does *not* compute the right-hand side of Eq. (1): no μ_j , no δ , no dissipation flux (their Eq. 48). The same note says so and gives three reasons the energy estimate would not carry over even if it did:

1. The papers reconstruct the *flux* by finite differences on a uniform grid (neither 2009 paper treats a non-uniform grid; the 2011 one makes the flux points non-uniform only near the boundaries and for a different reason). EXHALE reconstructs the primitive *state* in a finite-volume sense on a radial grid whose spacing varies by a factor of about 20 across the domain, with volume-coordinate stencil coefficients of its own (`weno3_geometry_coefficients`), and hands the two states to an HLLC Riemann solver.
2. The estimate for a system holds in characteristic form. EXHALE reconstructs component by component, on primitive variables, which is the case the paper says it cannot bound.
3. The equations are the nonlinear Euler system with sources, for which the papers themselves claim no proof.

6. Assessment

What ESWENO offers, in the order the papers establish it:

1. A proof, not an observation, that the semi-discrete operator does not create energy for linear hyperbolic problems, on any data, with no TVB assumption and no large-time-step constraint; and the converse statement that conventional WENO's symmetric part can have positive eigenvalues.
2. Design-order accuracy at smooth extrema with any number of vanishing derivatives, through the weights of Eq. (5) and the $O(\Delta\xi^2)$ floor. This part EXHALE already has.
3. A systematic route to any order, at 15 to 20 percent more cost and the same stencil.
4. Boundary closures that keep the estimate on a finite domain, where conventional WENO closures lose either order or stability.

What it does not offer: a proof for the nonlinear Euler equations, for a component-wise reconstruction, or for a finite-volume reconstruction on a stretched grid, which are the three properties of EXHALE's hydrodynamics.

For the questions open in this code the papers are therefore not a lever yet. The base sound-wave oscillation and the marching limit cycle (`docs/Update_EXHALE_stage1.md` items P44 and P54) were traced to the boundary treatment and the time integration, not to the reconstruction weights, and nothing measured on this code attributes an instability to a positive eigenvalue of the reconstruction's symmetric part. Adopting the dissipation term would first require restating the reconstruction as a finite-difference flux operator in characteristic variables on the uniform computational coordinate ξ (the grid is a smooth map $r(\xi)$, so that much is possible), and the finite-domain closures of Fisher et al. (2011) would replace the ghost-cell boundary treatment at both ends. That is a different hydrodynamic scheme, not an option to add to the present one, and it would be judged as any such change is: by the certification of the returned state and by the byte-identical regression on the cases whose scheme is unchanged. No such work is planned.

References

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